Vanguard (Defender Remake)

Author: Kristan Espinosa

Introduction

Defender is a highly popular arcade game released in 1981. In 1983 Defender was acknowledge as a highly difficult game with an average playtime of less than a minute (Softline, 1983, p.43). With a steep learning curve, the game can be frustrating or too difficult for some players. In games the concept of 'flow' involves fine-tuning the difficulty to match the player's skill level and game difficulty to prevent boredom or anxiety during playthrough (Csikszentmihalyi, 2008). A good flow can be achieved by starting the game at easy difficulty, with increasing difficulty at longer playthroughs (Baron, 2012).

Hence, the changes made in Vanguard aims to ease out the difficulty at the beginning stages of the game, to help accommodate new players, while providing an increasing challenge as the player further plays the game. New mechanics are also introduced to provide players with extra firepower to defend the humans against the invading enemies, while encouraging a high-risk and high-reward gameplay.

Changes

<u>Rescue</u>

The rescue mechanic is the main change introduced in Vanguard. It involves dropping the carried human to a drop off point located at the left side of the game space. Upon successful rescue, players gain score and increases their rescue count for player upgrades. This rescue mechanic implements a high-risk and high-reward gameplay. The player can only carry one human; thus, they need to strategically choose to either drop the human back to the ground or to carry the human to the station, the latter carrying a greater risk. If a human is successfully carried to the station, they are safe from abduction throughout the round. To balance this, if the player dies while carrying a human, not only do the player lose a life but the perished human counts towards the enemy's abduction count.

The purpose of this mechanic is to provide the player with rewards by successfully performing a high-risk task. This makes the gameplay more dynamic and incentivize players to constantly move around the game space. Therefore, the rescue mechanic enhances the original goal and gameplay of Defender.

<u>Friend-Ship</u>

The resultant action of the rescue mechanic is a positive feedback loop for the player. The Friend-Ship is an object in the game that provides extra firepower and upgrades for the player. For every three rescued human a Friend-Ship is created that follows the player. The player can only have three Friend-Ship at once. The first and second Friend-Ship will simultaneously shoot lasers with the player, thus enhancing firepower. The third Friend-Ship provides upgrades by halving the bomb's and shield's respawn requirement. To balance this mechanic, whenever a player loses a life one Friend-Ship is destroyed.

This mechanic is a significant change as it provides a positive feedback loop for the player. Rewarding the player following a high-risk action, is a great way to increase player interest (Schell, 2020, Rewards). As a result of this mechanic, players will feel more comfortable and confident with the game, thus facilitating a high-risk gameplay experience.

Human

The number of humans in Vanguard is reduced to five. This amount was chosen to balance the rescue and abduction count. These humans are evenly spread to balance out the rescue and abduction mechanics. Unlike in Defender, the human can safely, but slowly, drop itself towards the ground and can't be shot by the player. However, the player is free to drop off the human to the enemy station if they wish to increase the game's difficulty themselves.

Additionally, players can only carry one human at a time. Initially, the game was designed to allow multiple humans to be carried. However, after further testing, this rule can result to obtaining the Friend-Ship upgrade easily and quickly, while UFOs have less chance to perform abductions, thus affecting game balance. Furthermore, this defeated the purpose of the rescue mechanism on encouraging the player to move around the game space.

Player changes

Majority of the changes made with the player revolves around increasing the player's firepower and encouraging high-risk gameplay. The shield is a defensive mechanic that prevents life loss against projectiles and certain enemies. It nullifies a single mistake made by the player and helps with repositioning. Upon death, the player gets a three second invulnerability period giving them a chance to either reposition or destroy any surrounding enemies. Player starts off at three lives and yield an extra life for every 20,000 points, thus increasing the significance of the scoring system. Another change involves incrementing the fire rate of the player by picking up power-ups spawned randomly. A maximum of ten power-ups can be achieved and further power-ups increases the score by 1,000. The purpose of this mechanic is to create player progression to keep the player interested in the game. Moreover, collecting the power-up requires moving to a specific location, which involves player risk and reward. Power-ups are kept after death to prevent the feeling of hopelessness when the player die at harder rounds. Overall, these changes help with easing the learning curve while allowing for an aggressive playstyle.

Bomb

The bomb is a very powerful resource used to destroy multiple enemies at once. Its main purpose is to help the player reposition themselves when surrounded by enemies. Thus, one of the changes made was to null any points caused by the destruction of the bomb. Additionally, the player starts off with two bombs and can carry a maximum of three. The purpose of reducing the starting bomb count is for players to use them more strategically.

The method of restoring bomb count has also been changed. Bombs are restored by filling the yellow bar in the game's UI. This is simply achieved by moving a specific distance. Hence, this follows the main gameplay changes of Vanguard which is to encourage the player to play more actively and aggressively.

Abduction

The enemy UFOs abducts humans to the opposite side of the game space. This will increase the abduction count which builds up to a swarm event. For every five humans abducted, a swarm event will occur. The swarm event consists of three waves, with each waves containing different types of enemies. In addition, the UFOs receives upgrades based on abduction count. At three abduction count, the active UFOs gain a speed boost to increase the chances of a

successful abduction. At six abduction count, all UFOs receives an upgrade of shooting a projectile towards the player.

Chasing the abducted human to the enemy base while avoiding or destroying enemies along the way increases challenge and excitement in the game. In addition, the consequence of an abduction is a negative feedback or punishment for the player. It acts to balance out the positive feedback loop provided by the rescue mechanic.

Enemies

There are four different types of enemies in Vanguard. The UFOs are similar to the Lander enemy type of Defender. The UFOs naturally move in a zigzag pattern and can randomly turn active to abduct a random human. The element of randomness makes the enemies behaviour unpredictable, thus creating unpredictable and challenging gameplay (Schell, 2020, "The Lens of Chance", para. 1). The asteroids are similar to the Defender's Pod. When asteroids are shot by the player, they release three smaller asteroids. In Vanguard, the asteroids move naturally and slowly. Thus, they act as dangerous hazards for the player limiting their movement. The third type of enemy is the Enemy-Ship. It patrols across the game space in a linear path. When the player is nearby, it hovers around the player and shoots a laser projectile. The final type of enemy is the Drone. It patrols around a specific game space and quickly chases the player when nearby. They self-destruct after a short period of time, or by colliding with the player, producing an area of explosion. This explosion can bypass the player's shield, thus balancing its defensive effects.

These enemies are spawned from the top of the game space, to simulate invasion. Harder enemy types appear at later rounds of the game. This helps with game progression to keep the player engaged throughout the game. The more difficult enemy types are spawned at later rounds, thus increasing the challenge at longer playtime.

Level

The game must start at lower difficulty to attract new players, while increasing the difficulty at later stages of the game to prevent lengthy playthrough (Ellis, 2004). Additionally, research suggested that higher difficulty correlates with higher player motivation, but extremely high difficulty can decrease player motivation (Strojny et al., 2023). This coincides with the flow theory, suggesting that a balance between skill and challenge increases player attention during gameplay (Michailidis et al., 2018). Furthermore, a game's playtime is important since a short playthrough can be frustrating while extended playthrough can lead to player boredom or fatigue (Schell, 2020, "Short vs. Long", para. 1).

Vanguard's level system aimed to fulfil these criteria while targeting a playtime length of about five minutes. The levels help segment the game's difficulty. It starts off easy by only spawning the UFO and asteroid enemies to prevent very short playtime as in the case of Defender. This reduces the learning curve, helping new players with familiarizing themselves with the game's mechanics and enemy behaviour, while forming strategies to survive longer. Additionally, the earlier stages allow the player to build up their firepower through the powerups and rescue mechanics, preparing themselves for the more challenging rounds. For each completed level the number of enemies and their spawn rates are increased, resulting to an increasing challenge. Additionally, the room wrapping feature of Defender was removed since base stations are

located at opposite sides of the game space. As an alternative, the room width increases each level, making defending and rescuing humans more difficult.

The game's speed involving enemy, player, and projectile speed increases by five percent per level. Speed increase also offers extra difficulty as better reaction time is required, thus enhancing game attention at later levels. This concept is similar to Tetris where the basic mechanic is unchanged, but the increasing game speed offers a challenge for the players.

Overall, the level mechanic of Vanguard helps achieve player flow state. It helps keep the player's attention and motivation while preventing lengthy playthroughs.

Conclusion

In conclusion the changes introduced in Vanguard eases out the difficulty at the early stages of the game. The new mechanics facilitates a high-risk gameplay, making the game more interesting.

References

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Resources

Sound effects: https://sfxr.me/

Music: Sci-Fi Atmospheric Music Pack (License in game folder).

Raiders of the dying worlds

Alien Ai Abduction

Download The Sci-Fi Music Tracks | GameDev.tv.

Tutorials

Gamemaker manual. https://manual.yoyogames.com/#t=Content.htm

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